

1. Banking Transaction (Single Inheritance)

```
import java.util.Scanner;
```

```
class Account {
    private String accountNumber;
    private String accountHolder;

    public Account() {
    }

    public Account(String accountNumber, String accountHolder) {
        this.accountNumber = accountNumber;
        this.accountHolder = accountHolder;
    }

    public String getAccountNumber() {
        return accountNumber;
    }

    public void setAccountNumber(String accountNumber) {
        this.accountNumber = accountNumber;
    }

    public String getAccountHolder() {
        return accountHolder;
    }

    public void setAccountHolder(String accountHolder) {
        this.accountHolder = accountHolder;
    }

    public void displayDetails() {
        System.out.println("\n--- Account Details ---");
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder: " + accountHolder);
    }
}

class SavingsAccount extends Account {
    private double interestRate;

    public SavingsAccount() {
        super();
    }

    public SavingsAccount(String accountNumber, String accountHolder, double interestRate) {
        super(accountNumber, accountHolder);
        this.interestRate = interestRate;
    }

    public double getInterestRate() {
        return interestRate;
    }
}
```

```

    public void setInterestRate(double interestRate) {
        this.interestRate = interestRate;
    }

    @Override
    public void displayDetails() {
        super.displayDetails();
        System.out.println("Interest Rate: " + interestRate + "%");
    }
}

public class Q1_BankingTransaction {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.println("--- Single Inheritance: Banking Transaction ---");

        System.out.print("Enter Account Number: ");
        String accNum = scanner.nextLine();

        System.out.print("Enter Account Holder Name: ");
        String name = scanner.nextLine();

        System.out.print("Enter Interest Rate (%): ");
        double interest = scanner.nextDouble();

        SavingsAccount savingsUser = new SavingsAccount(accNum, name, interest);

        savingsUser.displayDetails();

        scanner.close();
    }
}

```

```

[annkurrerr@archlinux Expt6]$ java Q1_BankingTransaction
--- Single Inheritance: Banking Transaction ---
Enter Account Number: 123456789
Enter Account Holder Name: Ankur Kunde
Enter Interest Rate (%): 10

--- Account Details ---
Account Number: 123456789
Account Holder: Ankur Kunde
Interest Rate: 10.0%

```

2. Device Hierarchy (Multilevel Inheritance)

```

import java.util.Scanner;

class Device {
    private String deviceName;

```

```

public Device() {
    System.out.println("1. Device (Base Class) default initialized.");
}

public Device(String deviceName) {
    System.out.println("1. Device (Base Class) initialized.");
    this.deviceName = deviceName;
}

public String getDeviceName() {
    return deviceName;
}

public void setDeviceName(String deviceName) {
    this.deviceName = deviceName;
}
}

class Computer extends Device {
    private String processor;

    public Computer() {
        super();
        System.out.println("2. Computer (Intermediate/Derived Class) default initialized.");
    }

    public Computer(String deviceName, String processor) {
        super(deviceName);
        System.out.println("2. Computer (Intermediate/Derived Class) initialized.");
        this.processor = processor;
    }

    public String getProcessor() {
        return processor;
    }

    public void setProcessor(String processor) {
        this.processor = processor;
    }
}

class Laptop extends Computer {
    private int batteryLife;

    public Laptop() {
        super();
        System.out.println("3. Laptop (Derived Class) default initialized.");
    }

    public Laptop(String deviceName, String processor, int batteryLife) {
        super(deviceName, processor);
        System.out.println("3. Laptop (Derived Class) initialized.");
        this.batteryLife = batteryLife;
    }
}

```

```

    }

    public int getBatteryLife() {
        return batteryLife;
    }

    public void setBatteryLife(int batteryLife) {
        this.batteryLife = batteryLife;
    }

    public void displaySpecs() {
        System.out.println("\n--- Laptop Specifications ---");
        System.out.println("Device Name: " + getDeviceName());
        System.out.println("Processor: " + getProcessor());
        System.out.println("Battery Life: " + batteryLife + " hours");
    }
}

public class Q2_DeviceHierarchy {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("--- Multilevel Inheritance: Device Hierarchy ---");
        System.out.print("Enter Device Name: ");
        String name = scanner.nextLine();

        System.out.print("Enter Processor Details: ");
        String processor = scanner.nextLine();

        System.out.print("Enter Battery Life (hours): ");
        int battery = scanner.nextInt();

        System.out.println("\nCreating Laptop Object:");
        Laptop myLaptop = new Laptop(name, processor, battery);

        myLaptop.displaySpecs();

        scanner.close();
    }
}

```

```

[annkurrerr@archlinux Expt6]$ java Q2_DeviceHierarchy
--- Multilevel Inheritance: Device Hierarchy ---
Enter Device Name: iPhone
Enter Processor Details: Logic board
Enter Battery Life (hours): 24

Creating Laptop Object:
1. Device (Base Class) initialized.
2. Computer (Intermediate/Derived Class) initialized.
3. Laptop (Derived Class) initialized.

--- Laptop Specifications ---
Device Name: iPhone
Processor: Logic board
Battery Life: 24 hours

```

3. Course Management (Hierarchical Inheritance)

```
import java.util.Scanner;
```

```

class Course {
    private String courseName;
    private double baseFee;

    public Course() {
    }

    public Course(String courseName, double baseFee) {
        this.courseName = courseName;
        this.baseFee = baseFee;
    }

    public String getCourseName() {
        return courseName;
    }

    public void setCourseName(String courseName) {
        this.courseName = courseName;
    }

    public double getBaseFee() {
        return baseFee;
    }

    public void setBaseFee(double baseFee) {
        this.baseFee = baseFee;
    }
}

```

```

    public double calculateCourseFee() {
        return baseFee;
    }

    public void displayCourseInfo() {
        System.out.println("Course Name: " + courseName);
        System.out.println("Calculated Total Fee: $" + calculateCourseFee());
    }
}

class OnlineCourse extends Course {
    private double platformFee;

    public OnlineCourse() {
        super();
    }
    public OnlineCourse(String courseName, double baseFee, double platformFee) {
        super(courseName, baseFee);
        this.platformFee = platformFee;
    }
    public double getPlatformFee() {
        return platformFee;
    }
    public void setPlatformFee(double platformFee) {
        this.platformFee = platformFee;
    }
    @Override
    public double calculateCourseFee() {
        return getBaseFee() + platformFee;
    }
}

class OfflineCourse extends Course {
    private double facilityFee;
    public OfflineCourse() {
        super();
    }
    public OfflineCourse(String courseName, double baseFee, double facilityFee) {
        super(courseName, baseFee);
        this.facilityFee = facilityFee;
    }
    public double getFacilityFee() {
        return facilityFee;
    }
    public void setFacilityFee(double facilityFee) {
        this.facilityFee = facilityFee;
    }
    @Override
    public double calculateCourseFee() {
        return getBaseFee() + facilityFee;
    }
}

public class Q3_CourseManagement {
    public static void main(String[] args) {

```

```

Scanner scanner = new Scanner(System.in);
System.out.println("--- Hierarchical Inheritance: Course Management ---");
System.out.println("Enter Online Course Name: ");
String onlineName = scanner.nextLine();
System.out.println("Enter Base Fee and Platform Fee (separated by space): ");
double onlineBase = scanner.nextDouble();
double onlinePlatform = scanner.nextDouble();
scanner.nextLine();
System.out.println("\nEnter Offline Course Name: ");
String offlineName = scanner.nextLine();
System.out.println("Enter Base Fee and Facility Fee (separated by space): ");
double offlineBase = scanner.nextDouble();
double offlineFacility = scanner.nextDouble();
Course course1 = new OnlineCourse(onlineName, onlineBase, onlinePlatform);
Course course2 = new OfflineCourse(offlineName, offlineBase, offlineFacility);
System.out.println("\n--- Online Course Details Output ---");
course1.displayCourseInfo();
System.out.println("\n--- Offline Course Details Output ---");
course2.displayCourseInfo();
scanner.close();
}
}

```

```

[annkurrerr@archlinux Expt6]$ java Q3_CourseManagement
--- Hierarchical Inheritance: Course Management ---
Enter Online Course Name:
Object oriented programming system
Enter Base Fee and Platform Fee (separated by space):
700 200

Enter Offline Course Name:
Automata Theory and Formal Languages
Enter Base Fee and Facility Fee (separated by space):
1500 300

--- Online Course Details Output ---
Course Name: Object oriented programming system
Calculated Total Fee: $900.0

--- Offline Course Details Output ---
Course Name: Automata Theory and Formal Languages
Calculated Total Fee: $1800.0

```

4. Payment System (Abstract Class)

```

import java.util.Scanner;

abstract class Payment {
    private double amount;

    public Payment() {
    }

    public Payment(double amount) {
    }
}

```

```

        this.amount = amount;
    }

    public double getAmount() {
        return amount;
    }

    public void setAmount(double amount) {
        this.amount = amount;
    }

    public abstract void processPayment();
}

class CreditCardPayment extends Payment {
    private String cardNumber;

    public CreditCardPayment() {
        super();
    }

    public CreditCardPayment(double amount, String cardNumber) {
        super(amount);
        this.cardNumber = cardNumber;
    }

    public String getCardNumber() {
        return cardNumber;
    }

    public void setCardNumber(String cardNumber) {
        this.cardNumber = cardNumber;
    }

    @Override
    public void processPayment() {
        System.out.println("Processing Credit Card Payment...");
        System.out.println("Debiting $" + getAmount() + " from Card Number ending in: "
            + cardNumber.substring(Math.max(0, cardNumber.length() - 4)));
        System.out.println("Payment Successful!\n");
    }
}

class UPIPayment extends Payment {
    private String upiId;

    public UPIPayment() {
        super();
    }

    public UPIPayment(double amount, String upiId) {
        super(amount);
        this.upiId = upiId;
    }
}

```

```

public String getUpiId() {
    return upiId;
}

public void setUpiId(String upiId) {
    this.upiId = upiId;
}

@Override
public void processPayment() {
    System.out.println("Processing UPI Payment...");
    System.out.println("Debiting $" + getAmount() + " via UPI ID: " + upiId);
    System.out.println("Payment Successful!\n");
}
}

```

```

public class Q4_PaymentSystem {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.println("--- Abstract Class: Payment System ---");

        System.out.print("Enter amount for Credit Card Payment: ");
        double cardAmount = scanner.nextDouble();
        scanner.nextLine(); // consume newline
        System.out.print("Enter Credit Card Number: ");
        String cardNo = scanner.nextLine();

        System.out.print("\nEnter amount for UPI Payment: ");
        double upiAmount = scanner.nextDouble();
        scanner.nextLine(); // consume newline
        System.out.print("Enter UPI ID: ");
        String upiIdStr = scanner.nextLine();

        Payment payment1 = new CreditCardPayment(cardAmount, cardNo);
        Payment payment2 = new UPIPayment(upiAmount, upiIdStr);

        System.out.println("\n--- Initiating Transactions ---");

```

```

        payment1.processPayment();
        payment2.processPayment();
        scanner.close();
    }
}

```

```

[annkurrrr@archlinux Expt6]$ java Q4_PaymentSystem
--- Abstract Class: Payment System ---
Enter amount for Credit Card Payment: 50000
Enter Credit Card Number: 123456789012

Enter amount for UPI Payment: 20000
Enter UPI ID: 387283545

--- Initiating Transactions ---
Processing Credit Card Payment...
Debiting $50000.0 from Card Number ending in: 9012
Payment Successful!

Processing UPI Payment...
Debiting $20000.0 via UPI ID: 387283545
Payment Successful!

```

5. Logger System (Final Class)

```
import java.util.Scanner;

final class Logger {

    public Logger() {
    }

    public void logMessage(String message) {
        System.out.println("[LOG] " + java.time.LocalDateTime.now() + " : " + message);
    }
}

public class Q5_LoggerSystem {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        Logger systemLogger = new Logger();

        System.out.println("--- Final Class: Logger System ---");
        System.out.print("Enter a system event to log (e.g., User logged in): ");
        String event1 = scanner.nextLine();

        System.out.print("Enter another system event (e.g., Database connected): ");
        String event2 = scanner.nextLine();

        System.out.println("\n--- Generating Logs ---");

        systemLogger.logMessage(event1);
        systemLogger.logMessage(event2);

        scanner.close();
    }
}
```

```
[annkurrerr@archlinux Expt6]$ java Q5_LoggerSystem
--- Final Class: Logger System ---
Enter a system event to log (e.g., User logged in): User logged in
Enter another system event (e.g., Database connected): Database connected

--- Generating Logs ---
[LOG] 2026-05-28T23:53:30.368584422 : User logged in
[LOG] 2026-05-28T23:53:30.376465092 : Database connected
```